ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION

O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2024

2 hours 30 minutes

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Evaluate $6 + 6 \times 4 - 1$. **[2]**

2. Evaluate $-10 - (-9)$. **[3]**

3. Work out 3/5 + 1/6. Give your answer as a mixed number if possible. **[3]**

4. Find 40% of 250. **[3]**

5. Solve $6x + 1 = 31$. **[3]**

6. Share $18 in the ratio 3 : 3. **[4]**

7. Find (a) the HCF and (b) the LCM of 36 and 16. **[4]**

8. Write 200000 in standard form. **[3]**

9. Simplify 5^2 × 5^3. Leave your answer as a power of 5. **[3]**

10. $n(A) = 13$, $n(B) = 10$ and $n(A \cap B) = 2$. Find $n(A \cup B)$. **[3]**

11. Expand and simplify $6(x + 1) - 1x$. **[3]**

12. Factorise completely $4x + 8$. **[3]**

13. Factorise $x^2 + 7x + 12$. **[4]**

14. Make h the subject of the formula $A = 2\pi r(r + h)$. Take π as a symbol. **[4]**

15. Solve $3x + 2 < 8$. **[3]**

16. Solve the simultaneous equations
$x + y = 7$
$x - y = -3$. **[4]**

17. The nth term of a sequence is $5 + 3(n - 1)$. Find the 8th term. **[3]**

18. In triangle ABC, angle A = 57° and angle B = 77°. Find angle C. **[3]**

19. A regular polygon has 5 sides. Find the size of each exterior angle. **[3]**

20. A right-angled triangle has shorter sides 9 cm and 12 cm. Calculate the hypotenuse. **[4]**

21. In right-angled triangle XYZ, angle X = 60° and hypotenuse XZ = 10 cm. Calculate XY, the side adjacent to 60°. **[4]**

22. A triangle has base 10 cm and perpendicular height 5 cm. Find its area. **[3]**

23. Calculate the area of a circle of radius 7 cm. Take π = 22/7. **[4]**

24. A cylinder has radius 7 cm and height 5 cm. Find its volume. Take π = 22/7. **[4]**

25. Find the gradient of the line joining (2, 3) and (5, 11). **[3]**

26. Find the midpoint of the line joining (2, 3) and (5, 11). **[3]**

27. A bag contains 7 red, 4 blue and 1 green beads. A bead is drawn at random. Find the probability that it is red. **[3]**

28. Find the mean of 11, 5, 4, 11, 7. **[3]**

29. Work out (4 2 ; 2 2) (4 ; 4) where ( ) denotes a 2×2 matrix times a column vector. **[4]**

30. Given u = (1 ; 3) and v = (2 ; 3), find 3u + v. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).